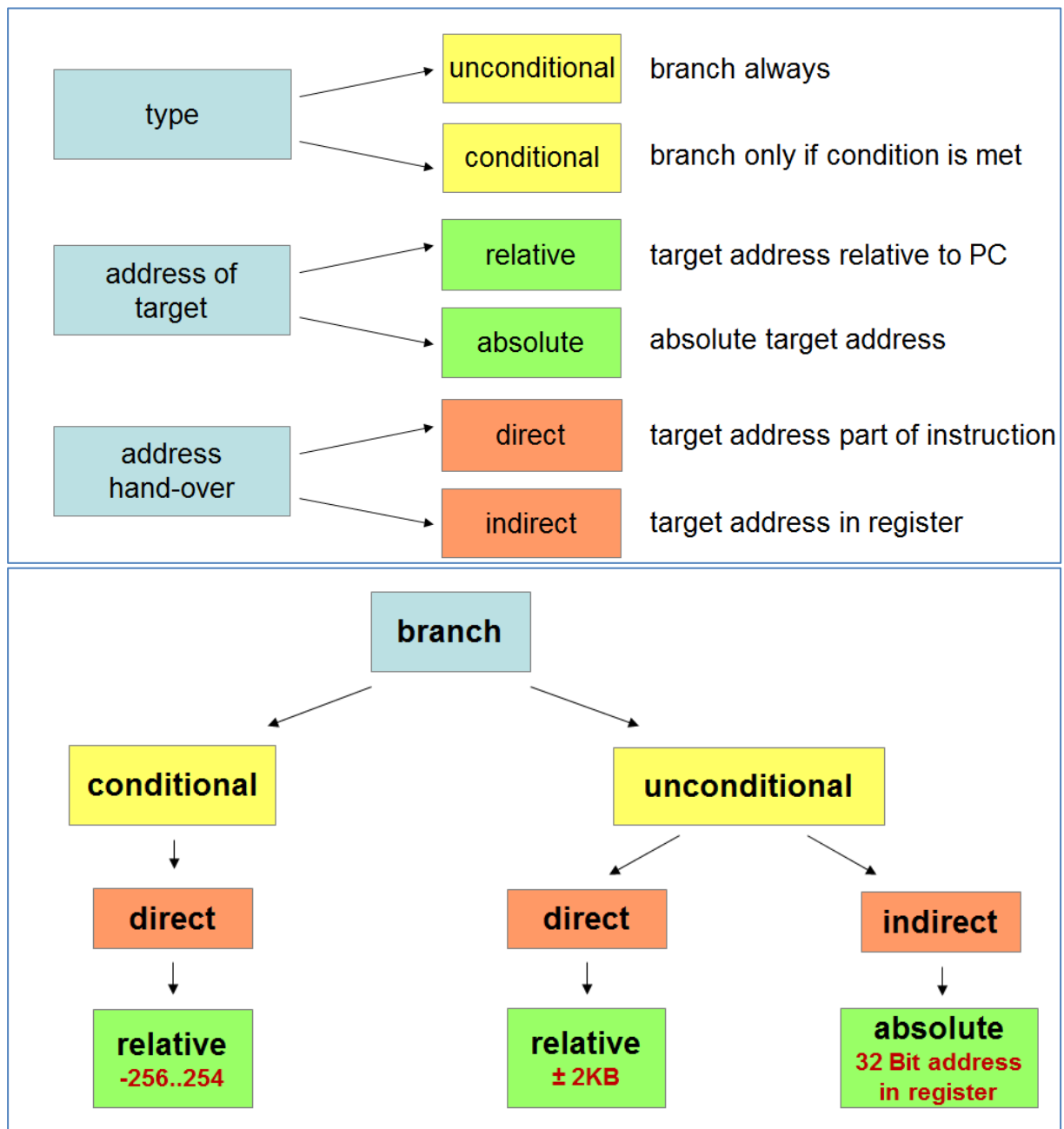


CT1 Exercises for Branching Instructions

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Exercise 1 – Unconditional Branches

The execution starts at line 10.

- 1) List the sequence of branch instructions that end in an infinite loop.
Do this by stating the branches in tabular form: *from – to*.
E.g. the first branch is **11 – 19** (branch unconditionally from line 11 to line 19).
- 2) At which line does the execution sequence finally loop forever?

10	label1:	ldr r0, =label5
11	label2:	bx r0
12	jumptable:	.word case0
13		.word case1
14		.word case2
15		.word case3
16		
17		.thumb @ Thumb code
18		.thumb_func
19	label5:	ldr r0, =label6
20		b label2
21		
22		.thumb
23		.thumb_func
24	label6:	ldr r2, =jumptable
25		adds r2, r2, #4
26		
27		.thumb
28		.thumb_func
29	label4:	ldr r2, [r2]
30		bx r2
31		
32		.thumb
33		.thumb_func
34	case0:	b case0
35		
36		.thumb
37		.thumb_func
38	case1:	ldr r2, =jumptable
39		movs r1, #3
40		lsls r1, r1, #2
41		adds r2, r2, r1
42		b label4
43		
44		.thumb
45		.thumb_func
46	case2:	b label1
47		
48		.thumb
49		.thumb_func
50	case3:	b case0

Your solution (the number of cells below is no hint)

1)

11-19								
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2)

Exercise 2 – Conditional Branches

The execution starts at line 10.

- 1) List which branch instructions jump to the given label.
Do this by stating the branches in tabular form: *from – to*.
- 2) What is the final value in R0 as hexadecimal value?

10		ldr r1, =0xffffffff-5	
11		ldr r2, =10	
12		ldr r3, =0x2341	
13			
14		movs r0, #0	@ mask of branches that are not taken
15			
16	label11:	adds r1, r1, #5	
17		bcs label12	
18		adds r0, r0, #0x01	@ set flag if no branch
19	label12:	adds r1, r1, #1	
20		bcs label13	
21		adds r0, r0, #0x02	@ set flag if no branch
22	label13:	ldr r5, =0x0f18c	
23		ands r1, r1, r5	
24		beq label21	
25		adds r0, r0, #0x04	@ set flag if no branch
26			
27	label21:	ldr r5, =2000	
28		subs r2, r2, r5	
29		bvs label22	
30		adds r0, r0, #0x08	@ set flag if no branch
31	label22:	ldr r5, =0x7fffffff	
32		subs r2, r2, r5	
33		bvs label23	
34		adds r0, r0, #0x10	@ set flag if no branch
35	label23:	adds r2, r2, #1	
36		bmi label31	
37		adds r0, r0, #0x20	@ set flag if no branch
38			
39	label31:	lsrs r3, r3, #1	
40		bcs endless	
41		adds r0, r0, #0x40	@ set flag if no branch
42			
43	endless	b endless	

Your solution (the number of cells below is no hint)

1)

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2) R0 = 0x.....

Exercise 3 – Comparison Instructions

The execution starts at line 10.

- 1) List which branch instructions jump to the given label.
Do this by stating the branches in tabular form: *from – to*.
- 2) What is the final value in R0 as hexadecimal value?

10		ldr r1, =0xFFFFFFFF	
11		ldr r2, =0x80000000	
12		ldr r3, =0X9CFA0000	
13		ldr r4, =0XC2350000	
14			
15		movs r0, #0	@ mask of branches that are not taken
16			
17	label11	cmp r1, #1	
18		blt label12	
19		adds r0, r0, #0x01	@ set flag if no branch
20	label12	cmp r1, #1	
21		blo label21	
22		adds r0, r0, #0x02	@ set flag if no branch
23			
24	label21	ldr r5, =0X7FFFFFFF	
25		cmp r2, r5	
26		bgt label22	
27		adds r0, r0, #0x04	@ set flag if no branch
28	label22	ldr r5, =0X7FFFFFFF	
29		cmp r2, r5	
30		bhi label31	
31		adds r0, r0, #0x08	@ set flag if no branch
32			
33	label31	ldr r5, =0X0040FFFF	
34		tst r3, r5	
35		bne label32	
36		adds r0, r0, #0x10	@ set flag if no branch
37	label32	tst r3, r5	
38		beq label41	
39		adds r0, r0, #0x20	@ set flag if no branch
40			
41	label41	ldr r5, =0x02100000	
42		tst r4, r5	
43		beq label42	
44		adds r0, r0, #0x40	@ set flag if no branch
45	label42	ldr r5, =0x10080000	
46		tst r4, r5	
47		beq endless	
48		adds r0, r0, #0x80	@ set flag if no branch
49			
50	endless	b endless	

Your solution (the number of cells below is no hint)

1)

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2) R0 = 0x.....

Solutions

Exercise 1:

- 1) 11 – 19, 20 – 11, 11 – 24, 30 – 38, 42 – 29, 30 – 50, 50 – 34, 34 – 34 ...
- 2) Loops at line 34

Exercise 2:

- 1) 20 – 22, 24 – 27, 33 – 35, 40 – 43
- 2) 0x29 (binary 0010'1001)

Exercise 3:

- 1) 18 – 20, 30 – 33, 35 – 37, 47 – 50
- 2) 0x66 (binary 0110'0110)